

ImageLab Progress Report Week 2

CMPS4191: Advanced Web Technologies

University of Belize

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<https://github.com/andreshungbz/imagelab>

IMAGELAB	Title	ImageLab Progress Report Week 2
	Course	CMPS4191 Advanced Web Technologies
	Semester	2026-1
	Due Date	Sep 16, 2026

Progress Summary

The second week involved implementing the single worker of the API server to perform the processing required to generate image variants.

Completed tasks included:

- Creation of the original image metadata and image job records after saving the original image to the server storage.
- Replaced Week 1 placeholder values for the HTTP 202 Accepted response.
- Prepared the routes for polling the image job (/v1/jobs/:id).
- Verified that only one worker was running.
- Created image manipulation helper functions to satisfy the current image variant requirements.
- Implemented image variant generation methods for the Image Variant model, storing image variant files to the server storage and image metadata records in the database.
- Verified job completed or failure by implementing a configured flag to simulate worker failure.
- Patched several bugs, such as in the UI and data service polling.
- Refactored various parts of the codebase, such as removing report-generation code, using UUIDv7 instead of UUIDv4 for server-controlled original filenames, and clearer error messages in the UI.
- Added a feature for dynamically displaying the header workspace badge.
- Refined the UI styling.

No requirements for the Week 2 Check-in were incomplete. Further testing and measurement are currently blocked until Week 3 and 4. Next in Week 3 will be completing the short polling and UI for the frontend. Much of the polling and UI foundation was already coded in Week 1, so the focus will be on testing and verification.

The one question for this week regards how to organize the server storage directories effectively. Currently, the original image and image variant records use big integers (as seen in the requirements). The current directory-based approach works, but we want to know whether there are better structures to use, or whether it is preferable to use UUIDs for the image and image-variant primary keys, as with job records.